

CHEATSHEET · QUICK REFERENCE

Lab 27 — Disaster Recovery: Replication and Site Failover

Compare hot/warm/cold/cloud sites, replicate async with rsync/lsyncd and sync with DRBD, plumb VIP failover, run a tabletop and match replication to the BIA.

Site types

Site	Time to bring up	Cost
Hot	Minutes	High
Warm	Hours	Medium
Cold	Days	Low
Cloud	Minutes-hours	Variable

Replication commands

```
rsync -avh --delete /srv/primary/ /srv/dr/
```

Async background replication

```
systemctl restart lsyncd
```

Near-real-time inotify replication

```
(DRBD / SRDF / semi-sync)
```

Synchronous - RPO 0, distance-bounded

Replication semantics

Term	Meaning
Async	Ack before replica has it (RPO = lag)
Sync	Blocks until both confirm (RPO 0)
App-consistent	Quiesce + snapshot before ship
Mirroring	One-way primary -> replica
Bidirectional	Both accept writes (conflict resolution)

DR test types + RPO match

Test / strategy	Detail
Tabletop	Walk the runbook (no risk)
Simulated failover	Isolated test env (low risk)
Live failover	Real traffic to DR (high risk)
lsyncd near-real-time	RPO ~10 s, medium cost
Sync block-level	RPO 0, high cost